

Predictors of Entry Into Medication-Assisted Therapy Based on Diagnoses of Opioid Dependence

Course: Statistics for Data Science Spring 2025

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Introduction

Opioids are a class of drugs developed to manage pain which also have significant potential to create a dependence (i.e., addiction), which can in turn have serious impacts on health, including death from overdose. In recent years in North America, a number of highly potent synthetic opioids, including fentanyl, have been circulating in the illegal drug supply leading to large increases in deaths from opioid overdose. In addition to the nature of the highly toxic and unpredictable illegal drug supply, overdose deaths also continue to persist due to barriers in accessing treatment services ([Government of Canada 2024](#)).

In both Canada and the United States, the recommended course of treatment for people with a dependence on opioids is to administer medication that is known as opioid agonist (or opioid replacement) therapy. This therapy mostly relies on administering opioids such as methadone or buprenorphine, which allow the patient to be weaned off of the opioids they are using in a more structured way ([CMA guidelines, 2018](#); [ASAM guidelines, 2015](#)).

Access to this treatment is not guaranteed. A quick review of the literature identifies the following key factors that can influence whether a person with an opioid dependence is treated with medication-assisted therapy:

- **Sociodemographic characteristics**, including income, health insurance status, race and ethnicity, and housing status (Landis et al., 2022, Krawczyk et al., 2017; Hochheimer & Unick, 2022, Gaeta Gazzola et al., 2023);
- **History and pattern of opioid use**, including strength/potency of opioids being used, duration, and frequency of use (Chang et al., 2023, Fernand, 2022);
- **Prior success or failure with treatment** (Chevez, 2024, Samples et al., 2018);
- **Patient knowledge of and perceptions of treatment** (Jaffe et al., 2024);
- **Health system and geographic access**, which can vary by province/state and for those living in rural or more resource-constrained areas (Nosyk et al., 2024, Levin et al., 2022);
- **Treatment retention**, which can be related to mental health and the use of other drugs in addition to opioids (Russell et al., 2025, Tsui et al., 2018).

For this project, a large open source dataset was used, the 2022 Treatment Episode Data Set Admissions (TEDS-A) from the United States' Substance Abuse and Mental Health Services Administration, to predict which patients admitted for an opioid dependence were given medication-assisted opioid therapy.

Data Preparation

The 2022 Treatment Episode Data Set Admissions initially presented as a significant dataset for analysis, yet as is the case in many data preparation efforts, the scope of the dataset was narrowed to draw reasonably bounded insights. There were four steps in the process. First, in Data Exploration, the relationships between a small selection of demographic parameters were examined with the goal of becoming familiar with the data. During Data Cleansing, the desired sampled dataset was selected and the missingness of the data was assessed. This was a key step in order to have a critical understanding of the models that we created. Third, this information about missingness was used to engineer necessary variables, so that models could be executed successfully. Last, multiple logistic regression models were executed. Each of these steps are explained below.

Data Exploration

The dataset has over 1 million rows and 62 columns and the first step was to get a high level understanding of this data. This was particularly important given that the majority of the group was not familiar with the domain area from which this dataset comes. All of the columns were named with a short-form code for what they truly represent and time was spent to list out the precise name of the column, in our dataset. The simple step of examining a dataframe with plain language column headers was done.

Next, the dataset was reduced to only opioid users by filtering on the DSM Diagnosis column. Per the TEDS-A-2022 documentation, this code identifies the reason for a client's encounter or treatment. When this column has a value of 5, the label is "Opioid dependence" and when this column has a value of 12, the label is Opioid abuse. There was no clarification in the documentation about the difference between Opioid dependence and Opioid abuse. Thus, the data was filtered to include both of these values, so as to optimize the size of the sampled dataset. This filtered dataset was 21% of the total dataset.

With this filtered dataset, the describe function for a dataframe was called. At this point the nature of the dataset became clear. The describe function did not yield meaningful data for mean, std, min, max, or quartiles. This was because all variables in the dataset were categorical. This important assessment set the direction for the data analysis completed for the rest of the project.

The exploration of the data then compared proportions of parameters between the sample and the total population, looking for insights to explore further. Interestingly, few differences were found in this exploration, suggesting that those admitted for treatment for an opioid dependence had similar characteristics to the entire population admitted for any type of substance use treatment. The ratio of male to female participants in the dataset was nearly identical for the sampled dataset and

the total population. Wait times across genders for sampled data and the total dataset were also assessed and again no particular outliers or noticeable differences in proportions were observed. As an example, 41.3% of females are admitted inside less than a week and 43.1% of males are admitted inside less than a week, for opioid users. Across the total dataset the numbers and the proportions are exceptionally similar (43.5% for women, 44.4% for men). However, during this step of the exploration the missingness of the data appeared. In exploring wait times by gender, 56.5% of the wait time data was missing for opioid users and 50.8% of the wait time data was missing across the entire dataset.

During the exploration of the data, the grouping of columns was also done, which carried through the model creation process. The sheer volume of columns was one complexity to contend with in wrangling the data. In addition, per column the number of possible identifiers was also lengthy and there was a risk that meaningful conclusions could not be drawn. So, for Wait Times, the categories were grouped, for ease of assessment.

The limitations of merely exploring the data developed – there is a table of Attendance at Support Groups from which meaningful conclusions could not be drawn. At this time, recognizing diminishing returns in data exploration was key.

The project turned to the narrower assessment of opioid users who also had medication-assisted opioid therapy. Per age range, the proportion of participants who had medication-assisted opioid therapy was examined relative to the participants who did not have medication-assisted opioid therapy. The proportion of participants who had medication-assisted opioid therapy increased with age, one of the first observations made about the data that would apply to the hypothesis. In this assessment, the missingness of the data remained approximately the same, in the 5-7% range.

Data Exploration is a useful exercise in moderation. Watching for diminishing returns is key, and turning to the desired dataset and leveraging regression analysis is the clearly the more powerful path forward.

Data Cleaning

It was necessary to narrow the sample set and ensure it was cleansed. This began by reducing the number of columns leveraged from 62 to 45. As models were developed, further column reduction was done, but at the outset the columns selected were ones thought to have a meaningful impact on opioid users who received medication-assisted therapy. The goal was to predict which patients admitted for an opioid dependence were given medication-assisted opioid therapy. The columns retained were demographic in nature - age, marital status, gender, race, etcetera. Additionally, the columns retained were related to drugs the participants may or may not reported at admission. Columns removed were primary,

secondary, tertiary used drugs. In addition, all patients with missing values for METHUSE – the dependent variable – were removed. The resulting dataset contained 296,907 rows.

The missing data was assessed. Using the seaborn heatmap function, patterns of missing data were visualized. This became a powerful determinant of subsequent data analysis. Missing data showed for Education, Marital Status, Referral Source, Recent Arrests, Employment Status, Veteran Status, Living Arrangement, Details on Primary Drug of Use at Admission, Recent attendance of substance use for self-help groups and health insurance status. Upon further analysis – leveraging a correlation of missing values between variables – confirmation was found that if one of the variables mentioned above is missing, it is highly likely that the other variables are also missing data. The key insight leveraged via the visualization is that the patterns of missingness appeared to be related to geographic location in the United States, as well as type of treatment service provider. This could be due to differing policies on sharing of information across different states and health institutions.

The missing data must be managed in order to run regression analysis. Considering that the data is categorical, there were three options considered to manage the missingness of the data:

1. **Mode Imputation** - Impute the most frequent value of the variable.
2. **Establish Missing as Category** – create a new category called “Missing” and fill NAs for this category
3. **Multivariate Imputation** – more complex statistical options.

The simple methods – option 1 and 2, were chosen, with a heavy use of option 1. This was done because the statistical complexities required to successfully leverage option 3 are beyond the scope of this course.

With this consideration for missingness, bar charts were leveraged. This allowed a quick assessment of categories under each parameter that had a significant majority of the data. An example is VET (Veteran Status). A significant majority of the VET data is populated with a 2 for “No”. This is demonstrated in the histogram quite clearly. For a column such as this, it would be reasonable to shift all missing data to the value of “No” in order to derive further insights into the data. All the variables/columns in the dataframe had this profile, except for three – Substance Use at Admission, Route of Admission, and Age at First Use. For all but these three variables, the SimpleImputer function from scikit-learn was leveraged to impute the most frequently occurring value into the data for the majority of the variables. For the remaining three variables and for any other columns still with missing data after running the SimpleImputer function, a Missing category was created.

Data Cleansing is always an important step in statistical analysis and this project was no exception. In particular, because the data was categorical, resolving missing data was a dominant aspect of data cleansing. The amount of missing data in the dataset was initially a concern but as the sample was narrowed, the relationships of missingness were assessed and the imputation function was leveraged, the data was in much better position for analysis.

Engineered and Reclassified Variables

With the notion that the data was cleansed, it was time to run models in order to analyze. In the early execution of this step, limitations in the results were observed and the strategy had to shift.

Specifically, a logistic regression was run with all NA values dropped. This dataset contained 76,617 entries, representing approximately a quarter of the 296K entries that exist for the full opioid addiction dataset. The X and Y values were selected and the newton method was used to run a logistical regression on the dataset. The pseudo R^2 value in this model was low, .472, which is not a sole indicator of fitness of the model but was not encouraging. As a comparable, the same model was run with the imputed values. Unfortunately, again, the pseudo R^2 value was 0.400. As a result, the strategy needed to expand; It was assessed that solely managing the missing data was an insufficient technique for drawing reasonable conclusions. With this assessment, engineering variables was the next step taken.

There were twelve variables selected for feature engineering, with a goal of creating more logical categories. In reducing the number of features overall in the model, the fitness of the model will likely improve and more intuitive conclusions could be drawn.

The Age and Race variables were reduced to 5 categories each. The Referral Source variable was reduced to 4 categories. The Education, Services, NoPriori, First Use, and Health Insurance variables were reduced to 3 categories. The Ethnic, Arrests, Frequency at Self Help Groups and Living Arrangement variables were reduced to 2 categories each. Reference is made to the notebook for these particulars – a sizeable aspect of the codebase and an important step for analysis. A dataframe was established for modelling, leveraging these engineered features, with the intent of drawing more meaningful conclusions while recognizing the bounds of feature engineering.

At this point in the data preparation, data had been explored, missing data managed and cleansed, models attempted without fruitful insights and feature engineering employed with the goal of having more meaningful conclusions. In order to focus the model generation, one last step was taken – a search of academic literature that suggests key determinants of initiating methadone/buprenorphine for opioid use disorder. These variables were: Race and Ethnicity, Income and Insurance Status,

Housing Status, Potency of Opioid Use, Duration & Severity of Use, History of Treatment Adherence, Lack of knowledge or misconceptions about treatment, Geographic Location, Psychosocial Stability. Using the crosstab function with MethUse, the values for these variables were visualized. The goal was an ease of confirmation on which variables had the most potential for meaningful conclusions. The final variables entered into the logistical regression were: Race, Ethnicity, Health Insurance, Living Arrangement, Route Of Admission, Frequency of Admission, Frequency at Self Help Groups, Division, Age, and Gender. While Gender did not show important differences in treatment with methadone/buprenorphine for opioid dependency, it was left in the model given its common sense use in reasoning about demographics.

Logistic Regression Models

Given the work completed for Data Preparation, a model was executed, and analysis was conducted on the variables mentioned above. Two steps were taken. First, the model was run repeatedly, assessing the results given the most recent variable added. Specifically, the model was first run with the Age variable, and the pseudo R^2 value was assessed to determine the impact of adding the new variable.

Unfortunately, the pseudo R^2 value did not yield significantly different results on the additional runs, except for Division. We address this in the analysis. A single block of code is provided in the notebook, where each variable was added and the model re-run. The second step taken was to run a correlation matrix looking for relationships between variables and then generating the odds ratio and 95% confidence intervals for variables. This was done to generate deeper insights into the variables selected, necessary for analysis given that the pseudo R^2 result of running the model successively with additional parameters did not yield abundantly meaningful results. During this step, a reference value was selected for each variable by not including it in the dataset. This gives meaning to the odds ratio – the value that the odds ratio is, is an increased (above 1.0) or decreased (below 1.0) likelihood of the value resulting in medication-assisted treatment, relative to the reference value.

To summarize, the Data Preparation phase was a rich and iterative process of exploration, cleansing, generating, and reshaping, in order to determine the most meaningfully accurate potential insights from the dataset. We then shifted our focus to analysis.

Analysis and Discussion

During model execution, an additive approach was taken, looking for the impact of each variable on the results with each additional variable addition. The pseudo R^2 value was leveraged and more impactfully, the odds ratio and 95% confidence intervals were leveraged. Division (geographical region) had the most significant

impact in this assessment. Each variable is reported below according to the larger category of variables as outlined in the Introduction:

SOCIODEMOGRAPHICS

Age: When adding in solely the Age variable, the pseudo R^2 value began at 0.010. The reference category used was the most prevalent age group: 30-39 years old. In calculating the finished models' odds ratios, age groups higher than the reference showed greater odds of medication-assisted treatment, while age groups lower than the reference had lower odds, reflecting the gradient of increased treatment by age group seen in the graph of age vs treatment. For example, for those aged 50 years and older, the odds ratio was 1.22, 95% CI [1.18, 1.27], meaning that older patients are 22% more likely to have received treatment with methadone/buprenorphine for opioid dependency as compared to 30-39 year olds.

Employment: When employment was added to the model the pseudo R^2 value remained at 0.010 and no additional information was gleaned from the finished models' odds ratio.

Frequency of Use: Adding Frequency of Use similarly had minimal impact, the pseudo R^2 value shifted to 0.014. The finished models' odds ratio indicated a minimal increase in likelihood of receiving treatment with the frequency of use variable added, with a value of 1.04, 95% CI [1.02, 1.05].

The results were generated in this fashion for Living Arrangement, Non Opioid Drugs Flag, Referral Source, and Race. With each addition, the pseudo R^2 value remained below 0.04 and the finished models' odd ratio generated no meaningful insights.

Latino ethnicity: In adding the variable ethnicity the pseudo R^2 value increased to 0.042. The reference source chosen was Non-Latino, meaning that all values for Non-Latino were dropped from the model to prevent multicollinearity from taking place. Upon inspection of the final model, Latino ethnicity, along with non-Latino ethnicity was dropped due to a high p-value of 0.61. Based on this, it can be stated that ethnicity between Latino and non-Latino categories were not statistically significant in assessing those likely to receive the methadone/buprenorphine for opioid dependency treatment.

HEALTH SYSTEM AND GEOGRAPHIC ACCESS

Health Insurance: The addition of health insurance had an impact on the dataset. The pseudo R^2 value increased to 0.06, which while still exceptionally small, relative to the other model runs, was a noticeable change. The reference value chosen was Medicaid. In the odds ratio, the value for No Insurance showed as 0.5, 95% CI [0.48, 0.52] meaning they were two times (i.e., $1/0.5$) less likely to receive medication-assisted therapy relative to those with Medicaid. Surprisingly, patients with private insurance were also relatively less likely to receive this type of treatment,

but not as noticeably – the odds ratio relative to the reference value was 0.70, 95% CI [0.65, 0.74]. We can reasonably assess that a lack of health insurance has a negative impact on a patient's opportunity to receive treatment with methadone/buprenorphine for opioid dependency or abuse. Unfortunately, this makes intuitive sense.

The Route of Admission and Self Help variables were added and yielded minimal impact on the data – no new meaningful results.

Census Division: The addition of the Division variable had a major impact. The pseudo R^2 value increased to 0.250, which was the most significant shift thus far. The reference value selected was US Territories and from the documentation, this specifies Puerto Rico as the detailed location. In the odds ratio calculations, relative to Puerto Rico (US Territories) the data variation is noticeable. Many of the northern and coastal areas of the US showed higher odds of treatment compared to Puerto Rico. The most extreme example was Middle Atlantic (New Jersey, New York, and Pennsylvania) which had an odds ratio of 7.26, 95% CI [6.81, 7.74], meaning that a patient from this region was over *seven times more likely* to receive treatment with methadone/buprenorphine for opioid dependency or abuse than someone living in Puerto Rico. One can consider that this is potentially about availability of the treatment; US Territories may not have the treatment option as readily available, given the distance of Puerto Rico, relative to the Middle Atlantic. On the other end of the spectrum, patients from the southern regions of the US had significantly lower odds of receiving medication-assisted treatment. For instance, the West South Central area (Arkansas, Louisiana, Oklahoma, and Texas) was exceptionally less likely to receive this treatment than people from the US Territories (Puerto Rico), with an odds ratio of 0.09, 95% CI [0.08, 0.10] (i.e. approximately ten times less likely). Patients from South Atlantic division (Delaware, District of Columbia, Florida, Georgia, Maryland, North Carolina, South Carolina, Virginia, and West Virginia) were also significantly less likely to receive this treatment relative to US Territories, with an odds ratio of 0.31, 95% CI [0.29, 0.33] and East South Central was similarly rated (0.33, 95% CI [0.30, 0.35]), comprised of Alabama, Kentucky, Mississippi, and Tennessee.

It was observed that the deeper south, the less likely the treatment was received relative to the US Territories. One can consider that these are ideological differences, knowing that the Deeper South is generally a more conservative region of the country where administering opioids to treat addiction may not be preferable for patients or their healthcare providers. However, it is also possible that these regions are more resource-constrained, and are unable to provide this type of treatment.

Frequency of First Use: The last variable considered was Frequency of First Use and this did not yield values that showed meaningful results.

Discussion and Conclusion

The results from the logistic regression model appear to largely reinforce what was established *a priori* in terms of what the scientific literature has to say about factors influencing medication-assisted therapy for people with an opioid dependency.

Sociodemographic factors such as race, health insurance coverage and housing status did have an influence in the model as did age and gender. It was unexpected that increasing age had such a strong influence while race and gender were less influential. Geography also had a strong impact on the model although it is unclear whether the differences between US states are due to resource constraints, larger rural populations, ideological differences in treatment approach, or some combination of these factors.

This study has some limitations in the availability of information to build the model since it is using an administrative dataset that was not necessarily built to provide all possible elements to answer the question of factors influencing medication-assisted treatment. For instance, as noted above, there is some information missing in terms of rural/urban status of patients and treatment locations, as well as health care resources by US state, attitudes around treatment, etc. It could be possible to build in some of this information using other data sources and also by using zip codes to join in data on further sociodemographic factors. A future study could try to address these gaps.

In conclusion, this analysis found a number of significant factors associated with medication-assisted treatment for people diagnosed with an opioid use dependency. In terms of the data quality, it was very high which facilitated data cleaning. However, there were missing data issues which were challenging to deal with. The logistic regression using exclusively categorical data also required some adaptation of approach since a more traditional linear regression with continuous variables has been the main focus of this course.

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